

5(a)	fructose ; I α or β	1
5(b)	<i>any two from:</i> formation of enzyme-substrate complex / substrate fits into active site ; A substrate binds to active site (so) reactants held close together (e.g. for bond formation) ; lowers, <u>activation energy</u> / <u>energy of activation</u> ; A E_a / A_e <i>can award annotated on a sketch</i> detail ; e.g. transfer of electrons strain on bonds alternative pathway holds substrate in a way that the bonds needed to be broken are exposed	2
5(c)	I <i>ref. to</i> denaturation, either mp2 or mp10 can be awarded <i>any four from:</i> <i>immobilised</i> 1 higher optimum temperature ; AW e.g. peak / max activity / 100% activity ; 2 74 / 75 / 76°C, v, 56 / 57 / 58 / 59°C ; A one correct value and calculation e.g. 17°C lower 3 <i>idea that</i> greater range of pH with, higher activity / greater stability ; AW <i>allow use of comparative data at one pH value to support if answer unclear</i> 4 <i>idea that</i> if pH changes, less effect on enzyme activity ; A resists changes in pH 5 <i>ref. to</i> more active at, higher temperatures / temperatures above 68°C ; A more thermostable / less effect on enzyme at higher temperatures A free enzyme not active above 70°C <i>allow use of comparative data at one temperature to support if answer unclear</i>	

Question	Answer	Marks
5(c)	6 one advantage of immobilised ; e.g. reusable ○ less / no, contamination of product (by enzyme) ○ easier, downstream processing / separation of enzyme from product ○ longer shelf-life useful if enzyme, difficult to obtain A <i>ref. to cost effective</i> ○ can use in continuous processes so more productive ○ suggestion that (can carry out process at) lower pH (more acid conditions and) may be more effective against microorganisms <i>free enzyme</i> 7 higher activity, at lower temperatures / between 35°C to 68°C ; <i>in context of a range, quoting one temperature not enough</i> 8 two temperatures, comparative values for both enzymes to support mp7 ; 9 optimum temperature lower so cheaper maintaining that temperature ; AW A lower temperatures cost efficient to maintain and enzyme works well / AW 10 (values of optimum) 74 / 75 / 76°C, v, 56 / 57 / 58 / 59°C ; A calculation, e.g. 17°C lower 11 lower optimum temperature still high enough to kill microorganisms ;	